Homework 2

Module 2 · Homework · No date

Name _____ Date _____

1. A 165-lb male received a single IV bolus dose of an antibacterial agent at a dose level of 20 mg/kg. The concentration of drug in the plasma was 113.7 mcg/mL and 49.5 mcg/mL at 2 and 8 hours, respectively, following administration of the dose. The antibacterial agent displays, linear, first-order, one compartment pharmacokinetics.

1a Module 2

What is the half-life of this agent in this patient?

1b Module 2

What is the expected initial plasma drug concentration of this dose of drug in this patient?

1c Module 2

What is the volume of distribution of this agent in this patient?

1d Module 2

How many hours following administration of the dose are required for 87.5% of the agent to be eliminated from the body?

1e Module 2

If the dose were administered at a level of 10 mg/kg, how many hours following administration of the dose are required for 87.5% of the drug to be eliminated from the body?

1f Module 2

What is the clearance of this agent in this patient?

2. The equation below represents the concentration of drug in the plasma following IV bolus administration of a 400 mg bolus dose. (Concentration is given in mcg/mL and time in hours)

$C_p = 9.46e^{-4,231t} + 6.25e^{-0.173t}$

2a Module 2

What is the elimination half-life of this agent?

2b Module 2

What is the initial concentration of drug in the plasma upon administration of the dose?

2c Module 2

What is the concentration of drug in the plasma 4 hours after administration of the dose?

2d Module 2

What is the apparent volume of distribution of the central compartment?